

# TMUA Formula + Fact Sheet

A compact but specification-aware reference sheet for TMUA Paper 1 and Paper 2. Use it for recall, but practise applying these under timed conditions.

**Exam reminder:** There is no formula booklet in TMUA. Paper 1 tests mathematical application. Paper 2 tests mathematical reasoning and proof, using Section 1 knowledge plus logic and argument skills. There is no negative marking, so attempt every question.

## 1. Algebra, indices, surds and polynomials

### INDICES

**Index laws:**  $a^m a^n = a^{m+n}$ ,  $\frac{a^m}{a^n} = a^{m-n}$

$$(a^m)^n = a^{mn}, (ab)^n = a^n b^n, a^{-n} = \frac{1}{a^n}$$

$$a^{1/n} = \sqrt[n]{a}, a^{m/n} = \sqrt[n]{a^m}$$

TMUA uses rational, negative and fractional indices frequently.

### SURDS

$$\sqrt{ab} = \sqrt{a}\sqrt{b}, \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$$

$$\sqrt{a^2 b} = a\sqrt{b} \text{ for } a \geq 0$$

**Conjugates:**  $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$

$$\frac{1}{a + \sqrt{b}} = \frac{a - \sqrt{b}}{a^2 - b}$$

Always simplify surds and rationalise denominators.

### QUADRATICS

For  $ax^2 + bx + c = 0$ :

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

**Discriminant:**  $\Delta = b^2 - 4ac$

$\Delta > 0$ : two real roots;  $\Delta = 0$ : repeated root;  $\Delta < 0$ : no real roots.

### COMPLETING THE SQUARE

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

For  $x^2 + bx + c$ :

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$$

Use for minimum/maximum value, vertex and inequalities.

### ROOTS

If  $ax^2 + bx + c = 0$  has roots  $\alpha, \beta$ :

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Vertex of  $y = ax^2 + bx + c$ :

$$x = -\frac{b}{2a}$$

### FACTORISATIONS

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$x^2 + bx + c = (x + p)(x + q) \text{ if } p + q = b, pq = c$$

### FACTOR AND REMAINDER THEOREM

**Factor theorem:**  $f(a) = 0 \iff (x - a)$  is a factor of  $f(x)$ .

**Remainder theorem:** remainder on division by  $(x - a)$  is  $f(a)$ .

For divisor  $ax + b$ , set  $ax + b = 0$ , so remainder is  $f(-b/a)$ .

### ALGEBRAIC DIVISION AND IDENTITIES

If  $f(x) = (x - a)q(x) + r$ , then  $r = f(a)$ .

An **identity** is true for all allowed values of  $x$ .

An **equation** is true only for its solutions.

Compare coefficients when two polynomial expressions are identical.

### SIMULTANEOUS EQUATIONS AND INTERSECTIONS

Line/quadratic method: substitute the linear expression into the quadratic, solve the resulting quadratic, then back-substitute.

Geometric meaning: solutions are intersection points of the two graphs.

A line can meet a quadratic in 0, 1, or 2 points; use the discriminant after substitution.

## 2. Inequalities, functions and graphs

### INEQUALITY RULES

If  $a < b$ , then  $a + c < b + c$ .

If  $c > 0$ , then  $ac < bc$ .

If  $c < 0$ , then  $ac > bc$ . **Reverse the sign.**

For products/quotients, use critical values and a sign chart.

### MODULUS

$|x| < a \iff -a < x < a$ , for  $a > 0$

$|x| > a \iff x < -a$  or  $x > a$ , for  $a > 0$

$|x - a|$  means distance from  $a$  on the number line.

### FUNCTIONS

$f(g(x))$  means apply  $g$  first, then  $f$ .

$\sqrt{x}$  means the **positive** square root.

One-to-one: each output comes from at most one input.

Many-to-one: different inputs can give the same output.

### QUADRATIC INEQUALITIES

To solve  $f(x) > 0$ , find the roots and identify where the graph is above the  $x$ -axis.

For  $a(x - r_1)(x - r_2)$ : outside the roots if  $a > 0$ ; inside the roots if  $a < 0$ .

Sketch first. It prevents most sign errors.

### AM-GM

For  $a, b \geq 0$ :

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Equality holds iff  $a = b$ .

Useful for minimum-value problems such as  $x + \frac{1}{x}$ .

### COMMON GRAPHS

- Line:  $y = mx + c$
- Quadratic:  $y = ax^2 + bx + c$
- Cubic: possible one, two, or three real roots
- Reciprocal:  $y = \frac{1}{x}$ , asymptotes  $x = 0, y = 0$
- Exponential:  $y = a^x, a > 0$
- Logarithm:  $y = \log_a x, x > 0$
- Modulus:  $y = |x|$

### GRAPH TRANSFORMATIONS

| New graph      | Transformation from $y = f(x)$             |
|----------------|--------------------------------------------|
| $y = f(x) + a$ | up $a$                                     |
| $y = f(x) - a$ | down $a$                                   |
| $y = f(x + a)$ | left $a$                                   |
| $y = f(x - a)$ | right $a$                                  |
| $y = af(x)$    | vertical stretch by factor $a$             |
| $y = f(ax)$    | horizontal stretch by factor $\frac{1}{a}$ |
| $y = -f(x)$    | reflection in the $x$ -axis                |
| $y = f(-x)$    | reflection in the $y$ -axis                |

### LOGARITHMS

$$a^b = c \iff b = \log_a c$$

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a(x/y) = \log_a x - \log_a y$$

$$\log_a(x^k) = k \log_a x$$

$$\log_a(1/x) = -\log_a x, \log_a a = 1$$

### EXPONENTIALS

$$y = a^x \iff x = \log_a y$$

$$\ln x = \log_e x$$

$$e^{\ln x} = x, \ln(e^x) = x$$

**extra** Change of base:  $\log_a x = \frac{\log_b x}{\log_b a}$

Change of base is useful but not required by the 2026 spec.

### 3. Sequences, series, binomial and counting

#### ARITHMETIC SEQUENCES

$$u_n = a + (n - 1)d$$

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l)$$

First difference is constant.

#### GEOMETRIC SEQUENCES

$$u_n = ar^{n-1}$$

Finite sum:

$$S_n = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1$$

Infinite sum:

$$S_\infty = \frac{a}{1 - r}, \quad |r| < 1$$

#### STANDARD SUMS

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

$$\sum_{k=1}^n c = cn$$

Useful in arithmetic series and proof-style questions.

#### RECURRENCE RELATIONS

A recurrence defines terms using previous terms, e.g.

$$x_{n+1} = f(x_n)$$

Generate terms carefully: start from the given  $x_1$  or  $x_0$ .

#### LINEAR AND QUADRATIC SEQUENCES

Linear sequence: constant first difference.

Quadratic sequence: constant second difference.

If  $u_n = an^2 + bn + c$ , then second difference  $= 2a$ .

#### BINOMIAL EXPANSION

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \binom{n}{n-r}$$

For coefficient of  $x^k$ , find the term where powers combine to  $x^k$ .

#### COMBINATIONS

$${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Use when order does **not** matter.

#### PERMUTATIONS

$${}^nP_r = \frac{n!}{(n-r)!}$$

Use when order **does** matter.

#### SYSTEMATIC COUNTING

If one task has  $m$  choices and the next has  $n$  choices, the ordered pair of tasks has  $mn$  choices.

Use lists, grids, tables and tree diagrams to avoid double-counting.

For repeated choices, check carefully whether repetition is allowed.

## 4. Coordinate geometry and circle geometry

### STRAIGHT LINES

Gradient:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

Point-gradient form:

$$y - y_1 = m(x - x_1)$$

General form:  $ax + by + c = 0$

Parallel:  $m_1 = m_2$

Perpendicular:  $m_1 m_2 = -1$

### DISTANCE AND MIDPOINT

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M = \left( \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

### CIRCLES: EQUATIONS

$$(x - a)^2 + (y - b)^2 = r^2$$

Centre  $(a, b)$ , radius  $r$ .

If  $x^2 + y^2 + 2gx + 2fy + c = 0$ :

$$\text{centre} = (-g, -f), \quad r = \sqrt{g^2 + f^2 - c}$$

### TANGENT TO CIRCLE

The tangent at a point is perpendicular to the radius at that point.

If radius gradient is  $m$ , tangent gradient is  $-1/m$ .

Find the radius gradient first, then use perpendicular-gradient rule.

### CIRCLE THEOREMS

- Angle at the centre is twice the angle at the circumference.
- Angle in a semicircle is  $90^\circ$ .
- Angles in the same segment are equal.
- Opposite angles in a cyclic quadrilateral sum to  $180^\circ$ .
- Tangent is perpendicular to radius at the point of contact.
- Angle between tangent and chord equals the angle in the alternate segment.
- Perpendicular from centre to chord bisects the chord.

### ANGLE FACTS

Angles on a straight line sum to  $180^\circ$ .

Angles around a point sum to  $360^\circ$ .

Vertically opposite angles are equal.

Triangle angles sum to  $180^\circ$ .

Exterior angle of triangle = sum of opposite interior angles.

### PARALLEL LINES

Corresponding angles are equal.

Alternate angles are equal.

Co-interior angles sum to  $180^\circ$ .

### POLYGONS

Sum of interior angles of  $n$ -gon:

$$(n - 2)180^\circ$$

Each interior angle of regular  $n$ -gon:

$$\frac{(n - 2)180^\circ}{n}$$

Sum of exterior angles:  $360^\circ$

### PYTHAGORAS

$$a^2 + b^2 = c^2$$

In 3D cuboid diagonal:

$$d = \sqrt{l^2 + w^2 + h^2}$$

## 5. Trigonometry, radians and geometry measurement

### RIGHT-TRIANGLE TRIG

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

### CORE IDENTITIES

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

### DOUBLE ANGLE

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$$

### SINE AND COSINE RULES

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

Area:

$$\text{Area} = \frac{1}{2} ab \sin C$$

Remember the ambiguous SSA case for sine rule.

### EXACT VALUES

| $\theta$      | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ |
|---------------|-----------|----------------------|----------------------|----------------------|------------|
| $\sin \theta$ | 0         | $\frac{1}{2}$        | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1          |
| $\cos \theta$ | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$        | 0          |
| $\tan \theta$ | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | undefined  |

### RADIANS

$$180^\circ = \pi \text{ radians}$$

$$\theta^\circ = \frac{\pi \theta}{180} \text{ radians}$$

$$\theta \text{ radians} = \frac{180\theta^\circ}{\pi}$$

### ARCS AND SECTORS

For  $\theta$  in radians:

$$\text{arc length} = r\theta$$

$$\text{sector area} = \frac{1}{2} r^2 \theta$$

$$\text{segment area} = \text{sector area} - \text{triangle area}$$

### TRIG GRAPHS

$\sin x, \cos x$ : period  $360^\circ$  or  $2\pi$   
 $\tan x$ : period  $180^\circ$  or  $\pi$   
 $\sin(-x) = -\sin x, \cos(-x) = \cos x$

### BEARINGS

Bearings are measured clockwise from North.  
 Always written with three figures, e.g.  $047^\circ, 230^\circ$ .  
 Draw North lines at every point involved.

### BASIC AREAS

Triangle:  $\frac{1}{2}bh$   
 Parallelogram:  $bh$   
 Trapezium:  $\frac{1}{2}(a+b)h$   
 Circle:  $\pi r^2$   
 Circumference:  $2\pi r = \pi d$

### VOLUMES

Cuboid:  $lwh$   
 Prism: cross-sectional area  $\times$  length  
 Cylinder:  $\pi r^2 h$   
 Sphere, cone and pyramid formulae are usually given if needed.

## 6. Differentiation and integration

### DERIVATIVE MEANING

$$f'(x) = \frac{dy}{dx}$$

Derivative = gradient of tangent = instantaneous rate of change.

Second derivative:  $f''(x) = \frac{d^2y}{dx^2}$

### POWER RULE

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Works for rational  $n$ , where expression is defined.

Simplify first, then differentiate.

### TANGENTS AND NORMALS

Tangent gradient at  $x = a$ :  $m = f'(a)$

Tangent:

$$y - y_1 = m(x - x_1)$$

Normal gradient:

$$m_{\text{normal}} = -\frac{1}{m}$$

### STATIONARY POINTS

Stationary point:  $f'(x) = 0$

If  $f''(x) > 0$ : local minimum.

If  $f''(x) < 0$ : local maximum.

If  $f''(x) = 0$ : inspect further.

### INCREASING/DECREASING

If  $f'(x) > 0$ , then  $f$  is increasing.

If  $f'(x) < 0$ , then  $f$  is decreasing.

Use sign charts for intervals.

### EXTRA DERIVATIVE RULES

**extra** Product:  $(uv)' = u'v + uv'$

**extra** Quotient:

$$\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$$

**extra** Chain:

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$$

### EXTRA STANDARD DERIVATIVES

**extra**  $\frac{d}{dx}e^x = e^x$

**extra**  $\frac{d}{dx}\ln x = \frac{1}{x}$

**extra**  $\frac{d}{dx}\sin x = \cos x$

**extra**  $\frac{d}{dx}\cos x = -\sin x$

### POWER INTEGRALS

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

For definite integrals:

$$\int_a^b f(x) dx = F(b) - F(a)$$

### EXTRA STANDARD INTEGRALS

**extra**  $\int \frac{1}{x} dx = \ln|x| + C$

**extra**  $\int e^x dx = e^x + C$

**extra**  $\int \cos x dx = \sin x + C$

**extra**  $\int \sin x dx = -\cos x + C$

### FUNDAMENTAL THEOREM

If  $F'(x) = f(x)$ , then:

$$\int_a^b f(x) dx = F(b) - F(a)$$

$$\frac{d}{dx} \left( \int_a^x f(t) dt \right) = f(x)$$

### COMBINING INTEGRALS

$$\int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

$$\int_b^a f(x) dx = -\int_a^b f(x) dx$$

### AREA VS INTEGRAL

Definite integral can be negative.

Area between curve and  $x$ -axis uses positive area.

If curve crosses the axis, split at roots.

### TRAPEZIUM RULE

For equal spacing  $h$ :

$$A \approx \frac{h}{2} [y_0 + y_n + 2(y_1 + \dots + y_{n-1})]$$

Over/underestimate depends on curvature.

### DIFFERENTIAL EQUATIONS

If  $\frac{dy}{dx} = f(x)$ , then:

$$y = \int f(x) dx$$

Use given point/condition to find  $C$ .

## 7. Vectors, transformations and similarity

### COLUMN VECTORS

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$$

$$k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$$

### VECTOR BETWEEN POINTS

From  $A(x_1, y_1)$  to  $B(x_2, y_2)$ :

$$\overrightarrow{AB} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Magnitude:

$$\left| \begin{pmatrix} a \\ b \end{pmatrix} \right| = \sqrt{a^2 + b^2}$$

### PARALLEL VECTORS

Vectors are parallel if one is a scalar multiple of the other:

$$\mathbf{a} = k\mathbf{b}$$

Collinear points often require parallel vectors.

### VECTOR PROOFS

To prove a quadrilateral is a parallelogram, show opposite sides are equal and parallel as vectors.

To prove midpoint, show position vector is average of endpoints.

$$\mathbf{m} = \frac{\mathbf{a} + \mathbf{b}}{2}$$

### SIMILARITY

Scale factor  $k$ :

Lengths scale by  $k$ .

Areas scale by  $k^2$ .

Volumes scale by  $k^3$ .

### CONGRUENCE

Triangle congruence tests:

SSS, SAS, ASA, RHS.

Congruent shapes have equal corresponding lengths and angles.

### TRANSFORMATIONS

- Translation: every point moves by the same vector.
- Reflection: mirror line is perpendicular bisector between point and image.
- Rotation: use centre, angle and direction.
- Enlargement: use centre and scale factor.
- Negative scale factor: enlargement plus reversal through centre.

### QUADRILATERALS

- Parallelogram: opposite sides parallel and equal.
- Rectangle: parallelogram with right angles.
- Rhombus: all sides equal.
- Square: rectangle and rhombus.
- Kite: two pairs of adjacent equal sides.
- Trapezium: one pair of parallel sides.

### 3D GEOMETRY VOCABULARY

Know: face, surface, edge, vertex, prism, cylinder, cone, sphere, hemisphere, pyramid, plan and elevation.

For 3D Pythagoras, first find a diagonal on one face, then use it in a second right triangle.

## 8. Number, units, ratio, percentage and bounds

### STANDARD FORM

$$a \times 10^n, \quad 1 \leq a < 10$$

For multiplication, multiply  $a$ 's and add powers of 10.

For division, divide  $a$ 's and subtract powers of 10.

### UNITS

Speed:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Density:

$$\rho = \frac{m}{V}$$

Pressure:

$$P = \frac{F}{A}$$

Convert units before calculating.

### PRIME FACTORISATION

Write numbers as products of prime powers.

HCF: take smaller powers common to both.

LCM: take larger powers appearing in either.

### RATIO

For ratio  $a : b$ , total parts =  $a + b$ .

First share:  $\frac{a}{a+b} \times \text{total}$

Second share:  $\frac{b}{a+b} \times \text{total}$

Scale factor =  $\frac{\text{new length}}{\text{old length}}$

### DIRECT PROPORTION

$$y \propto x \iff y = kx$$

$$y \propto x^n \iff y = kx^n$$

Find  $k$  using one known pair.

### INVERSE PROPORTION

$$y \propto \frac{1}{x} \iff y = \frac{k}{x}$$

$$y \propto \frac{1}{x^n} \iff y = \frac{k}{x^n}$$

### PERCENTAGES

$$\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$$

Increase by  $r\%$ : multiply by  $1 + \frac{r}{100}$ .

Decrease by  $r\%$ : multiply by  $1 - \frac{r}{100}$ .

### GROWTH AND DECAY

Compound growth:

$$A = P(1 + r)^n$$

Compound decay:

$$A = P(1 - r)^n$$

Here  $r$  is decimal rate, not percentage.

### BOUNDS

If  $x$  is rounded to nearest unit:

$$x - \frac{1}{2} \leq \text{true value} < x + \frac{1}{2}$$

Nearest tenth:  $\pm 0.05$

Nearest hundredth:  $\pm 0.005$

### RECURRING DECIMALS

Let  $x$  equal the recurring decimal.

Multiply by 10, 100, or 1000 to align recurring parts.

Subtract equations to remove recurrence.

### APPROXIMATION

Use sensible estimates before calculating. In no-calculator TMUA, approximation often eliminates answer choices.

Always track whether the question asks for exact value, approximation, upper bound, lower bound, or error interval.



## 9. Probability and statistics

### PROBABILITY SCALE

$$0 \leq P(A) \leq 1$$

Impossible: 0. Certain: 1.

Exhaustive outcomes sum to 1.

### ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If mutually exclusive:

$$P(A \cup B) = P(A) + P(B)$$

### INDEPENDENCE

Independent iff:

$$P(A \cap B) = P(A)P(B)$$

Equivalently:

$$P(A | B) = P(A)$$

### EXPECTED FREQUENCY

$$\text{expected frequency} = nP(A)$$

Relative frequency:

$$\frac{\text{number of successful outcomes}}{\text{number of trials}}$$

### AVERAGES

$$\text{mean} = \frac{\sum x}{n}$$

$$\text{range} = \text{max} - \text{min}$$

Median = middle value after ordering.

Mode = most frequent value.

### HISTOGRAMS

$$\text{frequency density} = \frac{\text{frequency}}{\text{class width}}$$

$$\text{frequency} = \text{frequency density} \times \text{class width}$$

### SCATTER GRAPHS

Positive correlation: as one variable increases, the other tends to increase.

Negative correlation: as one variable increases, the other tends to decrease.

**Correlation does not imply causation.**

Interpolation is within the data range; extrapolation is outside and less reliable.

### COMPLEMENT

$$P(A') = 1 - P(A)$$

Use "not" or "at least one" problems by complement.

### CONDITIONAL PROBABILITY

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

"Given B" means restrict the sample space to B.

### TREE DIAGRAMS

Multiply along branches.

Add separate routes to the same outcome.

Dependent events: second probability changes after first outcome.

Independent events: second probability stays the same.

### VENN COUNTING

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A'| = |U| - |A|$$

Fill the intersection first.

### GROUPED DATA

Estimated mean:

$$\bar{x} \approx \frac{\sum fx}{\sum f}$$

Use class midpoints for x.

Modal class = class with highest frequency.

### QUARTILES AND SPREAD

$$IQR = Q_3 - Q_1$$

Use median and IQR to compare skewed data sets.

## 10. Logic, language and proof

### LOGIC WORDS

"A and B" means both are true.

"A or B" means inclusive or: A, B, or both.

"Not A" means A is false.

TMUA tests precise reading of English.

### IF / ONLY IF / IFF

"If A, then B" means  $A \Rightarrow B$ .

"A if B" means  $B \Rightarrow A$ .

"A only if B" means  $A \Rightarrow B$ .

"A iff B" means both  $A \Rightarrow B$  and  $B \Rightarrow A$ .

### CONVERSE AND CONTRAPOSITIVE

Statement:  $P \Rightarrow Q$

Converse:  $Q \Rightarrow P$ , not equivalent in general.

Inverse:  $\neg P \Rightarrow \neg Q$ , not equivalent in general.

Contrapositive:  $\neg Q \Rightarrow \neg P$ , always equivalent.

### NECESSARY AND SUFFICIENT

If  $P \Rightarrow Q$ :

P is **sufficient** for Q.

Q is **necessary** for P.

"Fire implies oxygen": oxygen is necessary for fire.

### QUANTIFIERS

"For all" means every case.

"For some" means at least one case.

"There exists" means at least one example exists.

### NEGATING QUANTIFIERS

$$\neg(\forall x, P(x)) \iff \exists x, \neg P(x)$$

$$\neg(\exists x, P(x)) \iff \forall x, \neg P(x)$$

"Not all" means "at least one not".

### DE MORGAN

$$\neg(P \wedge Q) \iff \neg P \vee \neg Q$$

$$\neg(P \vee Q) \iff \neg P \wedge \neg Q$$

In English: not both = one or both fail.

### TRUTH-TABLE FACTS

**learning aid**  $P \Rightarrow Q$  is false only when P is true and Q is false.

A biconditional  $P \iff Q$  is true when P and Q have the same truth value.

### SET NOTATION

$\cup$ : union,  $\cap$ : intersection,  $\in$ : element of

$\subset$ : subset,  $\emptyset$ : empty set,  $A'$ : complement

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

### COUNTEREXAMPLES

To disprove "for all x,  $P(x)$ ", one counterexample is enough.

To disprove "there exists x such that  $P(x)$ ", prove no case works.

### DIRECT PROOF

Start from known assumptions and proceed logically:

$$A \Rightarrow B \Rightarrow C \Rightarrow \dots \Rightarrow Z$$

Each step must follow from earlier steps.

### PROOF BY CASES

Split into exhaustive cases, e.g. even/odd, positive/negative/zero.

Every possible case must be covered.

### PROOF BY CONTRADICTION

To prove S, assume  $\neg S$ .

Derive an impossibility.

Conclude S must be true.

### PROOF BY CONTRAPOSITIVE

To prove  $P \Rightarrow Q$ , prove:

$$\neg Q \Rightarrow \neg P$$

Often easier when Q contains a negative condition.

### INDUCTION

**extra** Base case: prove  $P(1)$ .

**extra** Inductive step: assume  $P(k)$ , prove  $P(k+1)$ .

**extra** Conclude  $P(n)$  for all required integers n.

### COMMON PROOF ERRORS

Invalid:  $ab = ac \Rightarrow b = c$  unless  $a \neq 0$ .

Invalid:  $\sin A = \sin B \Rightarrow A = B$  without checking other angle solutions.

Invalid: proving examples and claiming the general result.

### PAPER 2 PANIC CHECKLIST

- Underline "must", "could", "only if", "for all", "some", "not".
- Try small cases: 0, 1, 2, even/odd, positive/negative.
- Look for a counterexample before doing heavy algebra.
- For implications, test the contrapositive and the converse separately.
- No negative marking: eliminate, guess, flag, move.